
Script for the Video Demonstration

Introduction (30 seconds)

[Start the video with your face visible or a slide showing your name, course, and assignment title.]

Hello, my name is **Amin**, and this is my submission for the **COM745BigData** assignment on **Crime Data Analysis Using Hive and Zeppelin**.

The purpose of this project is to analyze publicly available crime data to identify trends and patterns. The dataset I used contains information on violent and property crimes, categorized by state and county.

I used **Hadoop (HDFS)** for storage, **Hive** for SQL-based querying, and **Zeppelin** for creating visualizations.

Step 1: Data Upload Verification (1 minute)

[Switch to your screen showing the HDFS directory in the terminal.]

The dataset was uploaded earlier to Hadoop's HDFS for distributed storage. Let's verify that the dataset is in the correct directory.

Here's the command to list the contents of the directory /user/hadoop/dataset/:

hdfs dfs -ls /user/hadoop/dataset/

[Run the command and highlight the uploaded file in the output.]

As you can see, the file 2017report.csv is successfully uploaded to HDFS. This dataset will now be queried using Hive.

Step 2: Data Cleaning in Hive (1 minute)

[Explain how the dataset was cleaned using the pre-created table crime_report_cleaned. Show verification queries.]

The dataset was cleaned earlier to handle missing values and standardize column types. Let's verify the cleaned data:

SELECT * FROM crime_report_cleaned LIMIT 10;

[Show the query results and highlight a few rows.]

Here, you can see the cleaned dataset where missing values in columns like Murder_and_manslaughter and Rape have been replaced with 0.

[Show a query to confirm no null values remain in the dataset.]

SELECT

  SUM(CASE WHEN Violentcrime IS NULL THEN 1 ELSE 0 END) AS null_violentcrime,

  SUM(CASE WHEN Murder_and_manslaughter IS NULL THEN 1 ELSE 0 END) AS null_murder,

  SUM(CASE WHEN Rape IS NULL THEN 1 ELSE 0 END) AS null_rape

FROM crime_report_cleaned;

[Explain the results.]

The output confirms that there are no null values in key columns, and the data is ready for analysis.

Step 3: Analysis in Hive (1.5 minutes)

[Run and explain key analysis queries in Hive.]

1. **Total Crimes by State:**
2. SELECT State, SUM(Violentcrime) AS TotalViolentCrime
3. FROM crime_report_cleaned
4. GROUP BY State
5. ORDER BY TotalViolentCrime DESC;

This query calculates the total violent crimes for each state. States with higher totals, like [highlight a state], require targeted policy interventions.

6. **Top 5 Counties with Violent Crimes:**
7. SELECT County, SUM(Violentcrime) AS TotalViolentCrime
8. FROM crime_report_cleaned
9. GROUP BY County
10. ORDER BY TotalViolentCrime DESC
11. LIMIT 5;

This query identifies the top 5 counties with the highest violent crime rates. These regions need focused resource allocation.

12. **Correlation Between Violent and Property Crimes:**
13. SELECT
14. SUM(Violentcrime) AS TotalViolentCrime,
15. SUM(Propertycrime) AS TotalPropertyCrime
16. FROM crime_report_cleaned;

This query provides the totals for violent and property crimes, which will help us analyze their correlation.

[Summarize the results briefly and transition to visualizations.]

Step 4: Visualizations in Zeppelin (1.5 minutes)

[Switch to Zeppelin and explain the visualizations created using Hive queries.]

1. **Bar Chart: Total Crimes by State**
    - **Query:**
    - SELECT State, SUM(Violentcrime) AS TotalViolentCrime
    - FROM crime_report_cleaned
    - GROUP BY State
    - ORDER BY TotalViolentCrime DESC
    - **Explanation:**
      "This bar chart shows the total violent crimes for each state. States like [highlight high-crime states] stand out as areas requiring significant attention."
2. **Scatter Plot: Correlation Between Violent and Property Crimes**
    - **Query:**
    - SELECT
    - SUM(Violentcrime) AS TotalViolentCrime,
    - SUM(Propertycrime) AS TotalPropertyCrime
    - FROM crime_report_cleaned

- **Explanation:**
  "This scatter plot demonstrates a positive correlation between violent and property crimes, meaning regions with higher violent crimes also tend to have higher property crimes."

3. **Line Graph: Crime Trends Over Time**
   - **Query:**
   - SELECT Year, SUM(Violentcrime) AS TotalViolentCrime
   - FROM crime_report_cleaned
   - GROUP BY Year
   - ORDER BY Year
   - **Explanation:**
     "This line graph shows trends in violent crimes over the years, helping us identify periods with increased or decreased crime rates."

[Show each chart for a few seconds and briefly explain the insights.]

Conclusion (30 seconds)

[Summarize the project and findings.]

In conclusion, this project provided insights into crime patterns using Hadoop, Hive, and Zeppelin:

- States and counties with the highest crime rates were identified.
- A strong correlation between violent and property crimes was observed.
- These insights can help policymakers allocate resources effectively and improve public safety.

Thank you for watching my demonstration.
